

Chemical Calculations Formula List

1. Mole concept formulas

- Number of moles:

$$n = \frac{m}{M}$$

where m = mass, M = molar mass.

- Number of particles:

$$N = nN_A$$

where $N_A = 6.022 \times 10^{23}$.

- Mass from moles:

$$m = nM$$

- Molar mass:

$$M = \frac{m}{n}$$

2. Stoichiometry formulas

- Mole ratio from balanced equation:

$$\frac{n_1}{v_1} = \frac{n_2}{v_2}$$

where v = stoichiometric coefficient.

- Mass relation in reactions:

$$m = \left(\frac{v \times M}{v'} \right) \times n$$

This is applied through balanced-equation mole conversion.

3. Percentage composition formulas

- Percent composition:

$$\% \text{ element} = \frac{\text{mass of element in 1 mole of compound}}{\text{molar mass of compound}} \times 100$$

- Percent purity:

$$\% \text{ purity} = \frac{\text{pure mass}}{\text{sample mass}} \times 100$$

4. Empirical and molecular formula

- Empirical formula from mass ratio:

$$\text{moles of each element} = \frac{\text{given mass}}{\text{atomic mass}}$$

- Molecular formula factor:

$$n = \frac{\text{molecular mass}}{\text{empirical formula mass}}$$

$$\text{Molecular formula} = (\text{Empirical formula}) \times n$$

5. Solution concentration formulas

- Molarity:

$$M = \frac{n}{V}$$

where V is volume in liters.

- Molality:

$$m = \frac{n}{\text{kg of solvent}}$$

- Normality:

$$N = \frac{\text{gram equivalents}}{V(\text{L})}$$

- Equivalent weight:

$$\text{Equivalent weight} = \frac{\text{molar mass}}{n\text{-factor}}$$

- Mole fraction:

$$X_A = \frac{n_A}{\sum n}$$

- Mass percent:

$$\% w/w = \frac{\text{mass of solute}}{\text{mass of solution}} \times 100$$

- Volume percent:

$$\% v/v = \frac{\text{volume of solute}}{\text{volume of solution}} \times 100$$

- Mass by volume percent:

$$\% w/v = \frac{\text{mass of solute in g}}{\text{volume of solution in mL}} \times 100$$

- ppm:

$$\text{ppm} = \frac{\text{mass of solute}}{\text{mass of solution}} \times 10^6$$

- ppb:

$$\text{ppb} = \frac{\text{mass of solute}}{\text{mass of solution}} \times 10^9$$

6. Dilution formulas

- Classical dilution:

$$M_1V_1 = M_2V_2$$

- For normality:

$$N_1V_1 = N_2V_2$$

7. Titration formulas

- Acid-base or redox equivalence:

$$N_1V_1 = N_2V_2$$

- General equivalence relation:

$$\text{equivalents of A} = \text{equivalents of B}$$

8. Gas calculation formulas

- Ideal gas equation:

$$PV = nRT$$

- Combined gas law:

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

- Density of gas:

$$d = \frac{PM}{RT}$$

- Molar mass from gas data:

$$M = \frac{dRT}{P}$$

- Dalton's law of partial pressures:

$$P_{\text{total}} = \sum P_i$$

- Partial pressure:

$$P_i = X_i P_{\text{total}}$$

- Graham's law:

$$\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}}$$

9. Thermochemistry formulas

- First law of thermodynamics:

$$\Delta U = q + w$$

- Enthalpy:

$$H = U + PV$$

- Enthalpy change:

$$\Delta H = \Delta U + \Delta(PV)$$

- Heat at constant pressure:

$$q_p = \Delta H$$

- Heat at constant volume:

$$q_v = \Delta U$$

- Heat capacity:

$$q = mc\Delta T$$

- Molar heat capacity:

$$q = nC\Delta T$$

- Hess's law:

$$\Delta H_{\text{reaction}} = \sum \Delta H_f^\circ(\text{products}) - \sum \Delta H_f^\circ(\text{reactants})$$

10. Entropy and Gibbs energy

- Entropy change:

$$\Delta S = \frac{q_{\text{rev}}}{T}$$

- Gibbs free energy:

$$G = H - TS$$

- Gibbs free energy change:

$$\Delta G = \Delta H - T\Delta S$$

- Relation with equilibrium constant:

$$\Delta G^\circ = -RT \ln K$$

11. Equilibrium formulas

- Equilibrium constant:

$$K_c = \frac{[\text{products}]^{\text{coeff}}}{[\text{reactants}]^{\text{coeff}}}$$

- Pressure equilibrium constant:

$$K_p = \frac{(P_{\text{products}})^{\text{coeff}}}{(P_{\text{reactants}})^{\text{coeff}}}$$

- Relation between K_p and K_c :

$$K_p = K_c (RT)^{\Delta n}$$

12. Acid-base formulas

- pH:

$$\text{pH} = -\log[H^+]$$

- pOH:

$$\text{pOH} = -\log[OH^-]$$

- Water ion product:

$$K_w = [H^+][OH^-]$$

- Relation:

$$\text{pH} + \text{pOH} = 14$$

at 25°C.

- Henderson-Hasselbalch equation:

$$\text{pH} = \text{p}K_a + \log \frac{[\text{salt}]}{[\text{acid}]}$$

13. Solubility and ionic product

- Solubility product:

$$K_{sp} = [M^{m+}]^a [A^{n-}]^b$$

for M_aA_b .

- Ionic product:

$$Q_{sp} = \text{actual concentration product}$$

14. Electrochemistry formulas

- EMF of cell:

$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$$

- Nernst equation:

$$E = E^\circ - \frac{RT}{nF} \ln Q$$

- At 25°C:

$$E = E^\circ - \frac{0.0591}{n} \log Q$$

- Free energy from cell potential:

$$\Delta G = -nFE$$

- Standard free energy:

$$\Delta G^\circ = -nFE^\circ$$

- Conductance:

$$G = \frac{1}{R}$$

- Resistivity:

$$\rho = \frac{RA}{l}$$

- Conductivity:

$$\kappa = \frac{l}{RA}$$

- Molar conductivity:

$$\Lambda_m = \frac{\kappa \times 1000}{C}$$

15. Colligative property formulas

- Relative lowering of vapor pressure:

$$\frac{p^0 - p}{p^0} = X_{\text{solute}}$$

- Elevation in boiling point:

$$\Delta T_b = K_b m$$

- Depression in freezing point:

$$\Delta T_f = K_f m$$

- Osmotic pressure:

$$\pi = CRT$$

- van't Hoff factor:

$$i = \frac{\text{observed colligative property}}{\text{calculated colligative property}}$$

16. Rate and kinetics formulas

- Average rate:

$$\text{Rate} = \frac{\Delta[\text{conc.}]}{\Delta t}$$

- First-order rate law:

$$\ln \frac{[A]_0}{[A]_t} = kt$$

- First-order half-life:

$$t_{1/2} = \frac{0.693}{k}$$

- Arrhenius equation:

$$k = Ae^{-E_a/RT}$$

- Arrhenius linear form:

$$\ln k = \ln A - \frac{E_a}{RT}$$

17. Spectroscopy formulas

- Beer-Lambert law:

$$A = \epsilon lc$$

- Absorbance relation:

$$A = \log \frac{I_0}{I}$$

18. Density and concentration links

- Density:

$$d = \frac{m}{V}$$

- Mass from density:

$$m = dV$$

19. Purity and assay formulas

- Percentage purity:

$$\% \text{ purity} = \frac{\text{pure substance}}{\text{sample mass}} \times 100$$

- Assay by titration:

$$\% \text{ assay} = \frac{\text{experimental content}}{\text{theoretical content}} \times 100$$

20. Quick master sheet

- $n = \frac{m}{M}$
- $N = nN_A$
- $M = \frac{n}{V}$
- $m = \frac{n}{\text{kg solvent}}$
- $M_1V_1 = M_2V_2$
- $N_1V_1 = N_2V_2$
- $PV = nRT$
- $P_{\text{total}} = \sum P_i$
- $E_{\text{cell}} = E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ}$
- $E = E^{\circ} - \frac{0.0591}{n} \log Q$
- $\Delta G = -nFE$
- $\text{pH} = -\log[H^+]$
- $K_w = [H^+][OH^-]$
- $\Delta T_b = K_b m$
- $\Delta T_f = K_f m$
- $\pi = CRT$
- $A = \epsilon lc$